

$$\text{MisclassificationRate}(v) = 1 - \max_{y \in L} p(y|v)$$

$$\text{Gini}(v) = 1 - \sum_{y \in L} p(y|v)^2$$

$$\text{entropy}(v) = - \sum_{y \in L} p(y|v) \log p(y|v)$$

$$\text{InfGain}(v, a) = \text{entropy}(v) - \sum_{u \in \text{children}(v, a)} \frac{|u|}{|v|} \text{entropy}(u)$$

$$\text{IntrinsicInformation}(v, a) = - \sum_{u \in \text{children}(v, a)} \frac{|u|}{|v|} \log \frac{|u|}{|v|}$$

$$\text{GainRatio}(v, a) = \frac{\text{InfGain}(v, a)}{\text{IntrinsicInformation}(v, a)}$$

Law of Bayes, using the Independence Assumption

The probability of an event H given evidence E is:

$$p(H|E) = \frac{p(E|H)p(H)}{p(E)} = \frac{\prod_{i=1}^n p(E_i|H)p(H)}{p(E)}$$

i.e. it is assumed that $p(E|H) = \prod_{i=1}^n p(E_i|H)$ (naive).

Using densities in the NB calculations –

from Witten & Elbe

Assumption: The values of a follow a Gaussian distribution^a (also: normal distribution), with density function:

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma_a} e^{-\frac{(x-\mu_a)^2}{2\sigma_a^2}}$$

where μ_a is the sample mean:

$$\mu_a = \frac{1}{|D_a|} \sum_{x \in D_a} x.a$$

σ_a is the sample standard deviation, and

$$\sigma_a^2 \text{ the sample variance: } \sigma_a^2 = \frac{1}{|D_a|-1} \sum_{x \in D_a} (x.a - \mu_a)^2$$

$$\text{RandIndex}(x, y) = \frac{\text{agree}_{11}(x, y) + \text{agree}_{00}(x, y)}{\text{agree}_{11}(x, y) + \text{agree}_{00}(x, y) + \text{disagree}_{10}(x, y) + \text{disagree}_{01}(x, y)}$$

0/1 LOSS

$$= \frac{1}{N} \sum_{i=1}^N \mathcal{I}(\hat{\mathbf{y}}^{(i)} \neq \mathbf{y}^{(i)})$$

$$\text{JaccardCoefficient}(x, y) = \frac{\text{agree}_{11}(x, y)}{\text{agree}_{11}(x, y) + \text{disagree}_{10}(x, y) + \text{disagree}_{01}(x, y)}$$

HAMMING LOSS

$$= \frac{1}{NL} \sum_{i=1}^N \sum_{j=1}^L \mathcal{I}[\hat{y}_j^{(i)} \neq y_j^{(i)}]$$

Optimization criterion for K-Means

$$\text{SSE} = \sum_{i=1}^K \sum_{x \in C_i} \text{dist}(x, c_i)^2$$

where c_i is the centroid of cluster C_i with $c_{ij} = \frac{\sum_{x \in C_i} x_j}{|C_i|}, j = 1 \dots |F|$

$$\text{cohesion}(X) = 1 - \frac{1}{|X|} \sum_{x \in X} d(x, \text{center}(X))$$

$$\text{separation}(X) = \frac{1}{|\zeta|-1} \sum_{Y \in \zeta, Y \neq X} d(\text{center}(X), \text{center}(Y))$$

$$\text{purity}(\xi, \varphi) = \sum_{i=1}^K \frac{|X_i|}{|D|} \text{clusPurity}(X_i, \varphi)$$

$$\text{clusPurity}(X_i, \varphi) = \max_{j=1 \dots L} \{p_{ij}\}$$

$$p_{ij} = \frac{|X_i \cap C_j|}{|X_i|}$$

For a cluster $X \in \zeta$ and data point $x \in X$:

- a : average distance of x to the other data points in X
- b : average distance of x to the other data points in $\cup_{Y \in \zeta, Y \neq X} Y$
- $\text{Silh}(x) = \begin{cases} 1 - \frac{a}{b} & \text{if } a < b \\ \frac{b}{a} - 1 & \text{if } a \geq b \end{cases}$

so that $\text{cluSilh}(X) = \frac{1}{|X|} \sum_{x \in X} \text{Silh}(x)$ and

$$\text{silhouette}(\zeta) = \frac{1}{|\zeta|} \sum_{X \in \zeta} \text{cluSilh}(X)$$

χ^2 for two categorical attributes

(Section 3.2.1.1)

Let A, B be two nominal (categorical) attributes with c , respectively r distinct values, in a dataset with m instances. We compute

$$\chi^2 = \sum_{i=1}^c \sum_{j=1}^r \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

where o_{ij} is the observed frequency of the joint event (A_i, B_j) , and

$e_{ij} = \frac{\text{count}(A=a_i) \times \text{count}(B=b_j)}{m}$ is the expected frequency.

We test against the hypothesis that A, B are independent (H_0) with $(c-1) \times (r-1)$ degrees of freedom.^a

Pearson correlation coeff

(Section 7.2.2.3)

measures the degree of linear correlation between two variables X, Y with measurements $\{x_i\}$ and $\{y_i\}$ and means $\bar{x}, \bar{y}$:

$$\rho(X, Y) = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{(\sum_i (x_i - \bar{x})^2) (\sum_i (y_i - \bar{y})^2)}}$$

Spearman correlation coeff

is the Pearson correlation coeff for the *ranks of the variables*, i.e.

$$r_s = \rho(R(X), R(Y)) = \frac{\text{Cov}(R(X), R(Y))}{\sigma_{R(X)} \sigma_{R(Y)}}$$

where X, Y are numerical variables and $R(X), R(Y)$ are their ranks.

$$\text{Cov}(A, B) = E((A - \bar{A})(B - \bar{B})) = \frac{1}{m} \sum_{i=1}^m (a_i - \bar{A})(b_i - \bar{B})$$